

AERO SENTINEL

AUTONOMOUS AIRSPACE ANOMALY
DETECTION SYSTEM

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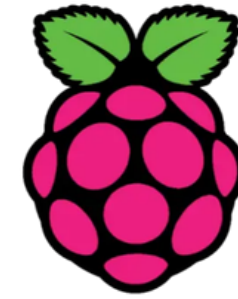
WHY ?

Currently, global aviation monitoring relies heavily on centralized cloud platforms (like FlightRadar24) and unencrypted ADS-B (Automatic Dependent Surveillance-Broadcast) radio signals. This creates significant vulnerabilities:

- Vulnerability to Spoofing: Unencrypted signals can be easily injected with fake data (Ghost flights).
- Single Points of Failure: During network outages or natural disasters, local authorities lose visibility of their airspace if cloud servers go down.
- Target Market: This project is designed for local aviation authorities, cybersecurity researchers, and emergency response teams who require an independent, resilient, and decentralized radar system.

THE SOLUTION

MAKE THE INVISIBLE VISIBLE



All processing runs locally on a Raspberry Pi 5.
No cloud required.

We propose an autonomous Edge AI IoT Node. By combining an SDR (Software Defined Radio) antenna with a Raspberry Pi 5, we create an independent radar station capable of capturing, decoding, and analyzing flight data locally. Instead of sending data to the cloud, an embedded Artificial Intelligence model evaluates the physical behavior of each aircraft in real-time, instantly flagging anomalies or emergency situations directly at the edge.

WHAT DOES THIS SOLUTION DO?

COLLECT

- Tunes into the 1090 MHz frequency to capture raw ADS-B radio waves from commercial and private aircraft.

DECODES

- Translates analog radio waves into structured, human-readable tabular data (JSON/CSV).

ANALYZE

- Feeds the flight data into an embedded Machine Learning model to evaluate the physical correlation between altitude, speed, and location.

ALERT

- Automatically detects physical anomalies (e.g., impossible nose-dives, spoofed coordinates) and displays them on a live local dashboard without requiring internet access.

DATA RECOLLECTION

To ensure the AI understands the specific dynamics of the local airspace, we conducted a rigorous data collection phase:

- Current Dataset: Approximately 260,000 real flight records.
- Timeframe: 7 days of continuous monitoring (April 8 - April 15, 2026).
- Strategy: We are expanding this database to capture seasonal, weather-related, and different daily traffic patterns to make the AI's training significantly more robust and reduce false positives.

HOW WE WILL TRAIN THE AI?

Instead of using Convolutional Neural Networks (CNNs) which are designed for image processing, our system relies on tabular numerical data. Therefore, we utilize Isolation Forest, an Unsupervised Machine Learning algorithm perfectly suited for anomaly detection.

- Why Isolation Forest? Instead of learning what a "normal" plane looks like (which takes massive computational power), it isolates outliers by building random decision trees. If a data point requires very few steps to be isolated, it is flagged as an anomaly.
- Features Evaluated: * Altitude_ft & Speed_kt: The core physics of flight dictate that planes fly faster at higher altitudes where the air is thinner. A plane flying at 650 knots at 500 ft breaks these physical laws and triggers an alert.
- Latitude & Longitude: Allows the model to implicitly learn standard commercial flight corridors. Deviations from these "invisible highways" are flagged.
- Data Cleaning: We heavily filter the dataset before training to remove extreme SDR sensor noise, forcing the AI to focus on subtle, tactical anomalies rather than mere radio static.

THE HARDWARE & SOFTWARE USE

HARDWARE

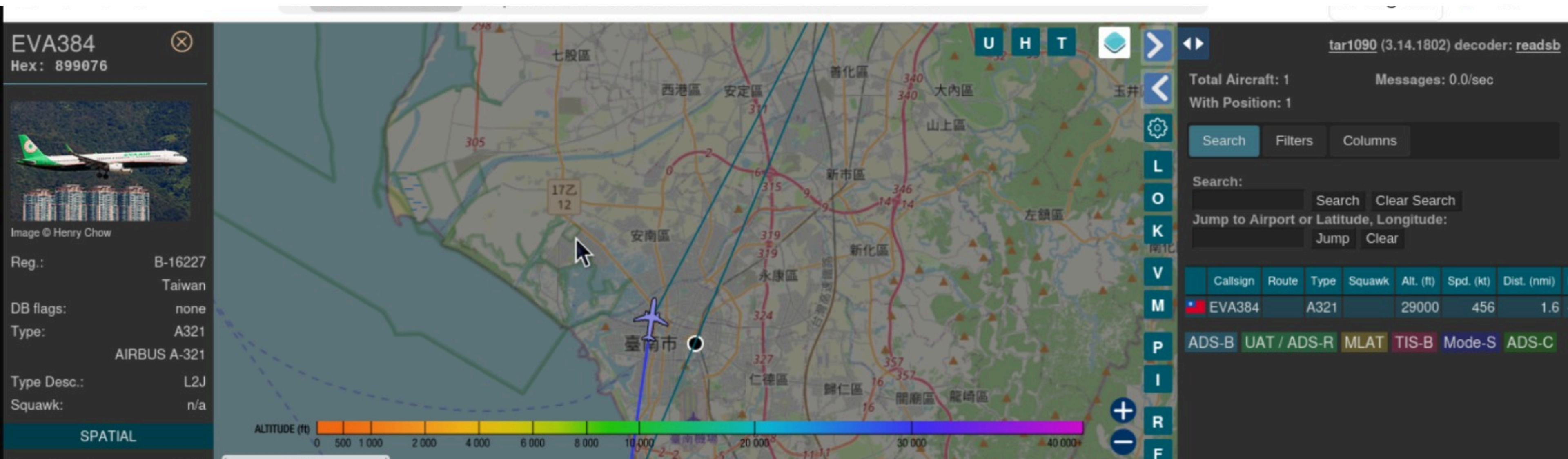
- RASPBERRY PI 5
- RTL-SDR V4 ANTENNA



SOFTWARE

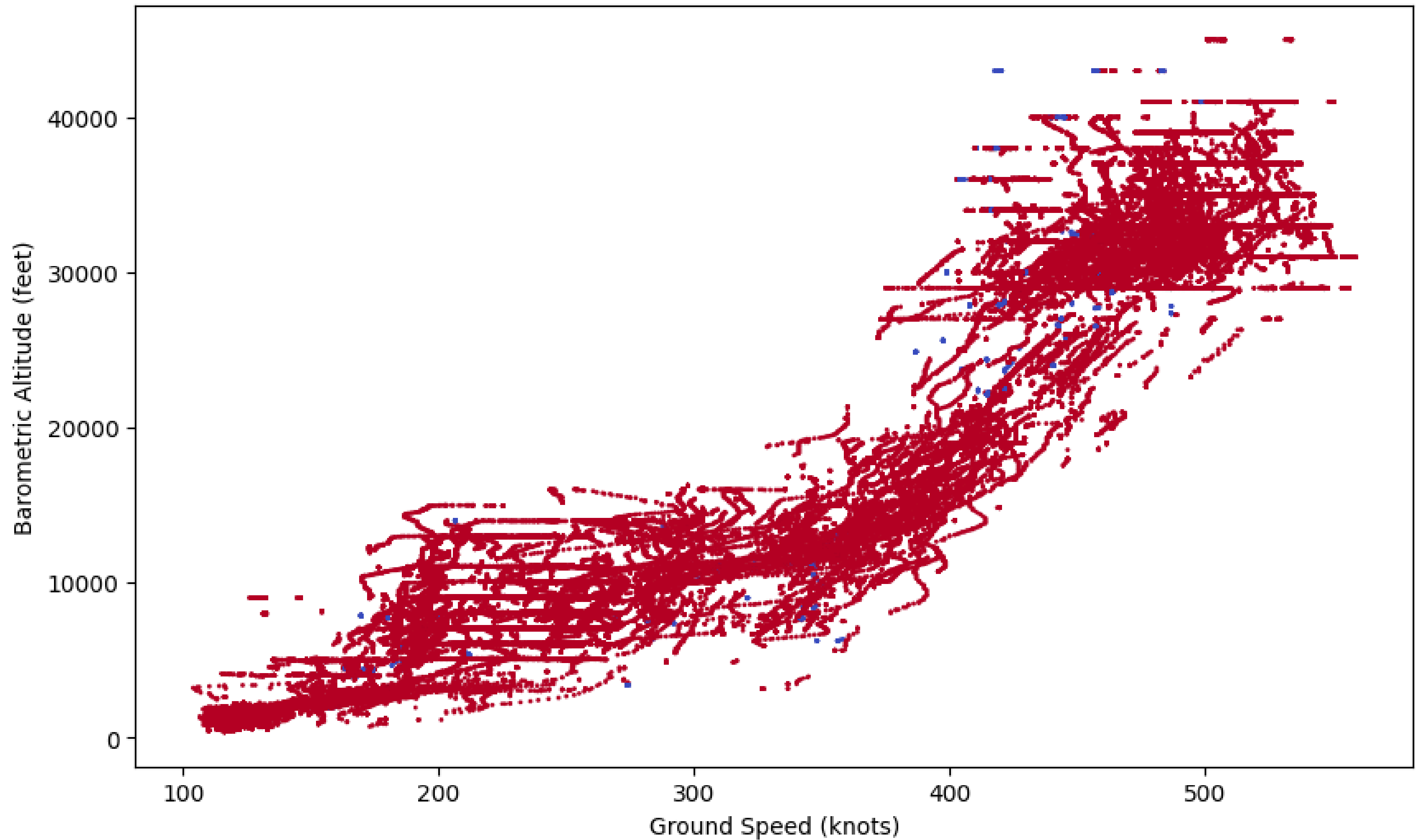
- READSB / DUMP1090: FOR DECODING RAW RF SIGNALS INTO JSON.
- PYTHON (PANDAS, SCIKIT-LEARN): FOR DATA PROCESSING, CLEANING, AND MACHINE LEARNING MODELING.
- GOOGLE COLAB: FOR HIGH-SPEED CLOUD TRAINING OF THE INITIAL DATASET.
- STREAMLIT: TO BUILD THE INTERACTIVE WEB DASHBOARD FOR THE IOT INTERFACE.

ADD ONS & IMPROVEMENTS



```
umEye/.venv/bin/python /home/porphyra/Documents/Programming_Projects/ADS-B/dataset_collector.py
[2026-04-08 16:06:05] → JJA2212 | Alt: 33000 ft | Vel: 466.0 kt | Pos: 22.874606, 120.165558
[2026-04-08 16:06:07] → JJA2212 | Alt: 33000 ft | Vel: 466.0 kt | Pos: 22.874606, 120.165558
[2026-04-08 16:06:09] → JJA2212 | Alt: 33000 ft | Vel: 466.0 kt | Pos: 22.874606, 120.165558
[2026-04-08 16:06:11] → JJA2212 | Alt: 33000 ft | Vel: 465.8 kt | Pos: 22.889374, 120.167908
[2026-04-08 16:06:13] → JJA2212 | Alt: 33000 ft | Vel: 465.8 kt | Pos: 22.893448, 120.168507
[2026-04-08 16:06:15] → JJA2212 | Alt: 33000 ft | Vel: 465.8 kt | Pos: 22.897751, 120.169206
[2026-04-08 16:06:17] → JJA2212 | Alt: 33000 ft | Vel: 465.8 kt | Pos: 22.902025, 120.169881
```


Airspace Anomaly Detection: Blue (Normal) vs Red (Anomaly)



THANK YOU

FOR YOUR ATTENTION

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